

Random Attention

every evictor scores the cache and keeps the top-K — we keep the prompt and drop the score

Score-based eviction

H2O · SnapKV · R-KV · VaSE · TriAttention



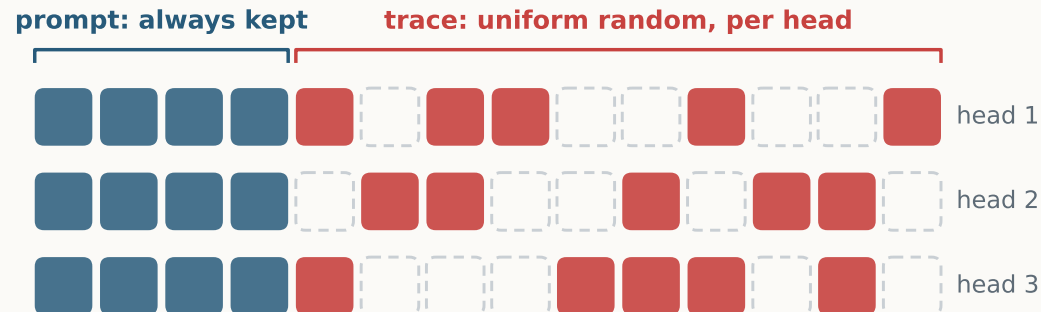
the question competes by score

TriAttention pins the whole prompt; the others keep only sinks

cost: a scoring pass at every eviction

Random Attention (ours)

no scores, no calibration, nothing to tune



every head keeps a different subset

so some copy of what the model still needs survives

cost: none — no scoring pass at all

matches the strongest evictor

4 models × 6 reasoning tasks

+32-43% tokens/s

vLLM, 32k-token generations

arxiv.org/abs/2609.03430

github.com/SalesforceAIResearch/Random-Attention